



Koneru Lakshmaiah Education Foundation

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SECTION : DBSE&DBD Y25 Section-8

MYSQL STORED PROCEDURES



MYSQL STORED PROCEDURES

Examples and Practice Exercises for MySQL Workbench

1. Setup: Employees Table

```
CREATE DATABASE IF NOT EXISTS employeedb1;
```

```
USE employeedb1;
```

```
CREATE TABLE IF NOT EXISTS employees (
```

```
    emp_id INT PRIMARY KEY,
```

```
    emp_name VARCHAR(50),
```

```
    dept_id INT,
```

```
    salary DECIMAL(10,2)
```

```
);
```

```
INSERT INTO employees VALUES
```

```
(101,'Rahul',1,50000),
```

```
(102,'Priya',2,65000),
```

```
(103,'Anil',1,55000),
```

```
(104,'Sneha',3,70000),
```

```
(105,'Kiran',2,48000);
```

2. Verify Data

Question: Display all employee records.

```
SELECT * FROM employees;
```

Procedure 1 – Display All Employees

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
198 COMMIT;
199 END $$
200
201 DELIMITER ;
202
203 • CALL TransferSalaryAmount(101,102,2000);
204
205 • SELECT * FROM employees
206 WHERE emp_id IN (101,102);
```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	53000.00
102	Priya	2	67000.00
NULL	NULL	NULL	NULL

result 90 Result 91 employees 92 Result 93 Result 94 employees 95 Result 96 Result 97 Result 98 Result 99 Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 453	12:44:41	CALL AddBonus(@bonus)	0 row(s) affected	0.000 sec
✓ 454	12:44:41	SELECT @bonus AS UpdatedBonus LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
⚠ 455	12:44:41	DROP PROCEDURE IF EXISTS TransferSalaryAmount	0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeedb1.Trans...	0.000 sec
✓ 456	12:44:41	CREATE PROCEDURE TransferSalaryAmount(IN fromEmp INT, I...	0 row(s) affected	0.016 sec
✓ 457	12:44:41	CALL TransferSalaryAmount(101,102,2000)	0 row(s) affected	0.000 sec
✓ 458	12:44:41	SELECT * FROM employees WHERE emp_id IN (101,102) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

169 END $$
170
171 DELIMITER ;
172
173 • SET @bonus = 10000;
174
175 • CALL AddBonus(@bonus);
176
177 • SELECT @bonus AS UpdatedBonus;

```

Result Grid

UpdatedBonus
15000.00

employees 78 Result 79 Result 80 employees 81 Result 82 Result 83 employees 84 Result 85 Result 86 Result 87 Read Only x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 403	12:43:59	SELECT @total AS TotalEmployees LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓ 404	12:43:59	DROP PROCEDURE IF EXISTS AddBonus	0 row(s) affected	0.015 sec
✓ 405	12:43:59	CREATE PROCEDURE AddBonus(INOUT bonus DECIMAL(10,2)) BEG...	0 row(s) affected	0.000 sec
✓ 406	12:43:59	SET @bonus = 10000	0 row(s) affected	0.000 sec
✓ 407	12:43:59	CALL AddBonus(@bonus)	0 row(s) affected	0.000 sec
✓ 408	12:43:59	SELECT @bonus AS UpdatedBonus LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

MYSQL x MYSQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

```

153 SELECT COUNT(*) INTO total
154 FROM employees;
155 END $$
156
157 DELIMITER ;
158
159 • CALL EmployeeCount(@total);
160
161 • SELECT @total AS TotalEmployees;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

TotalEmployees
5

employees 57 Result 58 Result 59 employees 60 Result 61 Result 62 employees 63 Result 64 Result 65 Result 66 Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 311	12:42:33	CREATE PROCEDURE SalaryStatus(IN eid INT) BEGIN DECLARE s...	0 row(s) affected	0.000 sec
✓ 312	12:42:33	CALL SalaryStatus(102)	1 row(s) returned	0.000 sec / 0.000 sec
⚠ 313	12:42:33	DROP PROCEDURE IF EXISTS EmployeeCount	0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeedb1.Empl...	0.000 sec
✓ 314	12:42:33	CREATE PROCEDURE EmployeeCount(OUT total INT) BEGIN SEL...	0 row(s) affected	0.016 sec
✓ 315	12:42:33	CALL EmployeeCount(@total)	1 row(s) affected	0.000 sec
✓ 316	12:42:33	SELECT @total AS TotalEmployees LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

138 SELECT 'High Salary' AS Status;
139 ELSE
140 SELECT 'Low Salary' AS Status;
141 END IF;
142 END $$
143
144 DELIMITER ;
145
146 • CALL SalaryStatus(102);

```

Result Grid

Status
High Salary

employees 48 Result 49 Result 50 employees 51 Result 52 Result 53 employees 54 Result 55 Result 56 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 270	12:40:16	DROP PROCEDURE IF EXISTS DepartmentAverageSalary	0 row(s) affected	0.000 sec
✓ 271	12:40:16	CREATE PROCEDURE DepartmentAverageSalary() BEGIN SELECT...	0 row(s) affected	0.015 sec
✓ 272	12:40:16	CALL DepartmentAverageSalary()	3 row(s) returned	0.000 sec / 0.000 sec
⚠ 273	12:40:16	DROP PROCEDURE IF EXISTS SalaryStatus	0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeedb1.Salar...	0.000 sec
✓ 274	12:40:16	CREATE PROCEDURE SalaryStatus(IN eid INT) BEGIN DECLARE s...	0 row(s) affected	0.000 sec
✓ 275	12:40:16	CALL SalaryStatus(102)	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

116 FROM employees e
117 JOIN departments d
118 ON e.dept_id = d.dept_id
119 GROUP BY d.dept_id, d.dept_name;
120 END $$
121
122 DELIMITER ;
123
124 • CALL DepartmentAverageSalary();

```

Result Grid

dept_name	EmployeeCount	AverageSalary
CSE	2	55000.000000
ECE	2	56500.000000
EEE	1	70000.000000

employees 40 Result 41 Result 42 employees 43 Result 44 Result 45 employees 46 Result 47 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
233	12:39:44	CALL EmployeeDepartment()	5 row(s) returned	0.000 sec / 0.000 sec
234	12:39:44	UPDATE employees SET salary = 55000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
235	12:39:44	SELECT * FROM employees WHERE emp_id = 101 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
236	12:39:44	DROP PROCEDURE IF EXISTS DepartmentAverageSalary	0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeedb1.Depa...	0.000 sec
237	12:39:44	CREATE PROCEDURE DepartmentAverageSalary() BEGIN SELECT...	0 row(s) affected	0.015 sec
238	12:39:44	CALL DepartmentAverageSalary()	3 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

98 DELIMITER ;
99
100 • CALL EmployeeDepartment();
101 • UPDATE employees
102   SET salary = 55000
103   WHERE emp_id = 101;
104
105 • SELECT * FROM employees
106   WHERE emp_id = 101;

```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	55000.00
* NULL	NULL	NULL	NULL

employees 33 Result 34 Result 35 employees 36 Result 37 Result 38 employees 39 x Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
199	12:37:50	INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,'E...)	0 row(s) affected, 3 warning(s): 1062 Duplicate entry '1' for key 'departm...	0.000 sec
200	12:37:47	CREATE DATABASE IF NOT EXISTS employeeedb1	1 row(s) affected, 1 warning(s): 1007 Can't create database 'employeeed...	0.000 sec
175	12:37:47	USE employeeedb1	0 row(s) affected	0.000 sec
176	12:37:47	USE employeeedb1	0 row(s) affected	0.000 sec
177	12:37:47	CREATE TABLE IF NOT EXISTS employees (emp_id INT PRIMARY...	0 row(s) affected, 1 warning(s): 1050 Table 'employees' already exists	0.016 sec
178	12:37:47	INSERT INTO employees VALUES (101,'Rahul',1,55000), (102,'Priya',2,...	Error Code: 1062. Duplicate entry '101' for key 'employees.PRIMARY'	0.000 sec

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

92 SELECT e.emp_name, d.dept_name, e.salary
93 FROM employees e
94 JOIN departments d
95 ON e.dept_id = d.dept_id;
96 END $$
97
98 DELIMITER ;
99
100 • CALL EmployeeDepartment();

```

Result Grid

emp_name	dept_name	salary
Rahul	CSE	75000.00
Priya	ECE	65000.00
Anil	CSE	55000.00
Sneha	EEE	70000.00
Kiran	ECE	48000.00

employees 27 Result 28 Result 29 employees 30 Result 31 Result 32 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 168	12:36:42	CALL AvgSalary()	1 row(s) returned	0.000 sec / 0.000 sec
⚠ 169	12:36:42	CREATE TABLE IF NOT EXISTS departments (dept_id INT PRIMA...	0 row(s) affected, 1 warning(s): 1050 Table 'departments' already exists	0.000 sec
⚠ 170	12:36:42	INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,'E...	0 row(s) affected, 3 warning(s): 1062 Duplicate entry '1' for key 'departm...	0.000 sec
✓ 171	12:36:42	DROP PROCEDURE IF EXISTS EmployeeDepartment	0 row(s) affected	0.015 sec
✓ 172	12:36:42	CREATE PROCEDURE EmployeeDepartment() BEGIN SELECT e.e...	0 row(s) affected	0.000 sec
✓ 173	12:36:42	CALL EmployeeDepartment()	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
92 SELECT e.emp_name, d.dept_name, e.salary
93 FROM employees e
94 JOIN departments d
95 ON e.dept_id = d.dept_id;
96 END $$
97
98 DELIMITER ;
99
100 CALL EmployeeDepartment();
```

Result Grid

emp_name	dept_name	salary
Rahul	CSE	75000.00
Priya	ECE	65000.00
Anil	CSE	55000.00
Sneha	EEE	70000.00
Kiran	ECE	48000.00

employees 27 Result 28 Result 29 employees 30 Result 31 Result 32 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 168	12:36:42	CALL AvgSalary()	1 row(s) returned	0.000 sec / 0.000 sec
⚠ 169	12:36:42	CREATE TABLE IF NOT EXISTS departments (dept_id INT PRIMA...	0 row(s) affected, 1 warning(s): 1050 Table 'departments' already exists	0.000 sec
⚠ 170	12:36:42	INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,'E...	0 row(s) affected, 3 warning(s): 1062 Duplicate entry '1' for key 'departm...	0.000 sec
✓ 171	12:36:42	DROP PROCEDURE IF EXISTS EmployeeDepartment	0 row(s) affected	0.015 sec
✓ 172	12:36:42	CREATE PROCEDURE EmployeeDepartment() BEGIN SELECT e.e...	0 row(s) affected	0.000 sec
✓ 173	12:36:42	CALL EmployeeDepartment()	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1

```
92 SELECT e.emp_name, d.dept_name, e.salary
93
94 JOIN departments d
95 ON e.dept_id = d.dept_id;
96 END $$
97
98 DELIMITER ;
99
100 CALL EmployeeDepartment();
```

Result Grid

emp_id	emp_name	dept_id
102	Priya	2

employees 27 Result 28 Result 29 employees 30 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 144	12:36:34	CALL EmployeeDepartment()	5 row(s) returned	0.000 sec / 0.000 sec
⚠ 145	12:36:40	CREATE DATABASE IF NOT EXISTS employeeedb1	1 row(s) affected, 1 warning(s): 1007 Can't create database 'employeeed...	0.031 sec
✓ 146	12:36:40	USE employeeedb1	0 row(s) affected	0.000 sec
✓ 147	12:36:40	USE employeeedb1	0 row(s) affected	0.016 sec
⚠ 148	12:36:40	CREATE TABLE IF NOT EXISTS employees (emp_id INT PRIMARY...	0 row(s) affected, 1 warning(s): 1050 Table 'employees' already exists	0.000 sec
✗ 149	12:36:40	INSERT INTO employees VALUES (101,'Rahul',1,50000), (102,'Priya',2,...	Error Code: 1062. Duplicate entry '101' for key 'employees.PRIMARY'	0.000 sec

MySQL Workbench

MYSQL x MYSQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```

77 • CREATE TABLE IF NOT EXISTS departments (
78     dept_id INT PRIMARY KEY,
79     dept_name VARCHAR(50)
80 );
81
82 • INSERT IGNORE INTO departments VALUES
83     (1, 'CSE'),
84     (2, 'ECE'),
85     (3, 'EEE');

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

AverageSalary
60600.000000

employees 16 Result 17 Result 18 employees 19 Result 20 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 110	12:36:01	SELECT * FROM employees WHERE emp_id = 101 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓ 111	12:36:01	DROP PROCEDURE IF EXISTS AvgSalary	0 row(s) affected	0.000 sec
✓ 112	12:36:01	CREATE PROCEDURE AvgSalary() BEGIN SELECT AVG(salary) AS...	0 row(s) affected	0.000 sec
✓ 113	12:36:01	CALL AvgSalary()	1 row(s) returned	0.000 sec / 0.000 sec
✓ 114	12:36:01	CREATE TABLE IF NOT EXISTS departments (dept_id INT PRIMA...	0 row(s) affected	0.047 sec
✓ 115	12:36:01	INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,E...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

MySQL Workbench

MySQL x MySQL x Local MySQL x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

```
68 • CREATE PROCEDURE AvgSalary()
69 BEGIN
70     SELECT AVG(salary) AS AverageSalary
71     FROM employees;
72 END $$
73
74 DELIMITER ;
75
76 • CALL AvgSalary();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

AverageSalary
59600.000000

employees 11 Result 12 Result 13 employees 14 Result 15 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 84	12:35:28	CREATE PROCEDURE IncreaseSalary(IN eid INT, IN amount DE...	0 row(s) affected	0.015 sec
✓ 85	12:35:28	CALL IncreaseSalary(101,5000)	1 row(s) affected	0.016 sec
✓ 86	12:35:28	SELECT * FROM employees WHERE emp_id = 101 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
⚠ 87	12:35:28	DROP PROCEDURE IF EXISTS AvgSalary	0 row(s) affected, 1 warning(s): 1305 PROCEDURE employeeedb1.AvgS...	0.000 sec
✓ 88	12:35:28	CREATE PROCEDURE AvgSalary() BEGIN SELECT AVG(salary) AS...	0 row(s) affected	0.000 sec
✓ 89	12:35:28	CALL AvgSalary()	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
48 • CREATE PROCEDURE IncreaseSalary(  
49     IN eid INT,  
50     IN amount DECIMAL(10,2)  
51 )  
52 BEGIN  
53     UPDATE employees  
54     SET salary = salary + amount  
55     WHERE emp_id = eid;  
56 END $$  
57  
58 DELIMITER ;  
59  
60 • CALL IncreaseSalary(101,5000);  
61  
62 • SELECT * FROM employees  
63 WHERE emp_id = 101;
```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	55000.00

Form Editor

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Navigator Query 1 x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report
- Performance Schema

Administration

Information

No object selected

Object Info Session

Query 1

```

34
35 • CREATE PROCEDURE GetEmployeeById(IN eid INT)
36 BEGIN
37     SELECT * FROM employees
38     WHERE emp_id = eid;
39 END $$
40
41 DELIMITER ;
42
43 • CALL GetEmployeeById(102);

```

Result Grid

emp_id	emp_name	dept_id	salary
102	Priya	2	65000.00

employees 4 Result 5 Result 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
39	12:33:25	DROP PROCEDURE IF EXISTS GetEmployees	0 row(s) affected	0.016 sec
40	12:33:25	CREATE PROCEDURE GetEmployees() BEGIN SELECT...	0 row(s) affected	0.016 sec
41	12:33:25	CALL GetEmployees()	5 row(s) returned	0.000 sec / 0.000 sec
42	12:33:25	DROP PROCEDURE IF EXISTS GetEmployeeById	0 row(s) affected, 1 warning(s): 1305 PROCEDURE empl...	0.000 sec
43	12:33:25	CREATE PROCEDURE GetEmployeeById(IN eid INT) B...	0 row(s) affected	0.016 sec
44	12:33:25	CALL GetEmployeeById(102)	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Navigator Query 1 x

MANAGEMENT

- Server Status
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- Data Import/Restore

INSTANCE

- Startup / Shutdown
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- Options File

PERFORMANCE

- Dashboard
- Performance Report
- Performance Schema

Administration

Information

No object selected

```
19 • DROP PROCEDURE IF EXISTS GetEmployees;
20
21 DELIMITER $$
22
23 • CREATE PROCEDURE GetEmployees()
24 BEGIN
25     SELECT * FROM employees;
26 END $$
27
28 DELIMITER ;
29
30 • CALL GetEmployees();
```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

Object Info Session employees 2 Result 3 x

Read Only

MySQL Workbench

MySQL x MySQL x Local Mysql x

File Edit View Query Database Server Tools Scripting Help

Navigator Query 1 x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report
- Performance Schema

Administration

Information

No object selected

Query 1

```
1 • CREATE DATABASE IF NOT EXISTS employeeedb1;
2 • USE employeeedb1;
3 • USE employeeedb1;
4 • CREATE TABLE IF NOT EXISTS employees (
5     emp_id INT PRIMARY KEY,
6     emp_name VARCHAR(50),
7     dept_id INT,
8     salary DECIMAL(10,2)
9 );
10
11 • INSERT INTO employees VALUES
12 (101, 'Rahul', 1, 50000.00);
13 (102, 'Priya', 2, 65000.00);
14 (103, 'Anil', 1, 55000.00);
15 (104, 'Sneha', 3, 70000.00);
16 (105, 'Kiran', 2, 48000.00);
```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00
* NULL	NULL	NULL	NULL

Insert new row

Object Info Session employees 1 x

Apply Revert